

```

1  --s16_2
2
3  LIBRARY ieee;
4  USE ieee.std_logic_1164.ALL;
5  USE ieee.numeric_std.ALL;
6
7  ENTITY s16_3_1b IS
8      PORT (
9          clk : IN STD_LOGIC;
10         u : IN STD_LOGIC;
11         x : OUT STD_LOGIC_VECTOR(1 DOWNTO 0)
12     );
13 END ENTITY;
14
15 ARCHITECTURE behavioral OF s16_3_1b IS
16     SIGNAL q : STD_LOGIC_VECTOR(1 DOWNTO 0) := (OTHERS => '0');
17     -- SUBTYPE counter_type IS INTEGER RANGE 0 TO 3;
18     SIGNAL count : UNSIGNED(1 DOWNTO 0) := "00";
19 BEGIN
20
21     count_p : PROCESS (clk)
22     BEGIN
23         IF rising_edge(clk) THEN
24             IF u = '1' THEN
25                 count <= count + 1;
26             ELSE
27                 count <= count;
28             END IF;
29         END IF;
30     END PROCESS;
31
32     -- output
33     x <= STD_LOGIC_VECTOR(count);
34
35 END ARCHITECTURE behavioral;

```